

CONSTRUCTION PROJECT COST & PERFORMANCE ANALYSIS

Data Analytics Project Report

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Project focus
Analysis of 1,000 construction projects covering budgets, actual costs, cost components, delays, contractors, project types, locations, project status, and cost relationships.

Analysis Stack

Data	CSV construction project dataset
Analysis	Python • Pandas • NumPy
Visualization	Matplotlib • Plotly
BI Dashboard	Power BI • DAX • Slicers
Business Output	KPIs • Trends • Comparisons • Insights

1. Executive Summary

The notebook documents an end-to-end construction analytics workflow. It loads a 1,000-row, 22-column dataset, inspects structure and descriptive statistics, calculates financial and operational KPIs, segments projects by type/location/contractor/status, analyzes cost components and delays, examines relationships between numeric measures, and supports dashboard visualization.

Metric	Result
Total projects	1,000
Total estimated budget	₹20,429.85 Cr
Total actual cost	₹20,804.38 Cr
Budget variance	₹374.52 Cr
Overall budget overrun	1.83%
Under-budget projects	456
Over-budget projects	544
Average delay	45.54 days
Maximum delay	150 days
Delayed projects	795
On-time projects	205

2. Data Understanding & Validation

- Loaded Construction_Project_Analytics.csv into a Pandas DataFrame.
- Verified the dataset shape as 1,000 rows × 22 columns.
- Inspected column names, data types, and non-null counts with df.info().
- Used df.describe() to review count, mean, standard deviation, minimum, quartiles, and maximum for numeric variables.
- The notebook output shows all 22 columns with 1,000 non-null values.

3. Financial Analysis

3.1 Budget Overview

Calculated total estimated budget, average estimated budget, total actual cost, budget variance, and overall budget overrun percentage.

The total estimated budget is ₹20,429.85 crore, while total actual cost is ₹20,804.38 crore. The resulting budget variance is ₹374.52 crore, corresponding to an overall overrun of 1.83%.

3.2 Budget Status Classification

Projects were separated into under-budget and over-budget groups by comparing Actual_Cost_INR with Estimated_Budget_INR. The notebook reports 456 under-budget projects and 544 over-budget projects.

3.3 Project-Type Budget Overrun

Project Type	Average Overrun
Commercial	1.80%
Industrial	1.96%
Infrastructure	1.68%
Residential	2.71%

The notebook's project-type comparison shows the highest average budget overrun for Residential projects at 2.71%.

4. Location Analysis

Location	Actual Cost (₹ Cr)
Ahmedabad	9,025.78
Gandhinagar	2,043.06
Mumbai	1,800.96
Pune	1,781.99
Rajkot	2,203.37
Surat	2,181.72
Vadodara	1,767.49

Location-wise analysis groups actual construction cost by city. The notebook reports the values above; total cost should be interpreted alongside the number of projects in each location.

5. Construction Cost Component Analysis

Component	Cost (₹ Cr)	Share
Material	8,282.25	39.81%
Labour	4,996.94	24.02%
Equipment	2,317.34	11.14%
Other	5,207.84	25.03%

The notebook calculates both absolute cost and percentage contribution of Material, Labour, Equipment, and Other costs to total actual cost.

6. Delay & Schedule Analysis

- Identified delayed projects using Delay_Days > 0.
- Reported 795 delayed projects and 205 on-time projects.
- Calculated an average delay of 45.54 days.
- Calculated a maximum observed delay of 150 days.
- Sorted projects by delay to identify the projects with the highest recorded delays.

7. Contractor Analysis

Contractor-level aggregation calculates total projects, average budget overrun, average delay, and total actual cost. The notebook output identifies MetroBuild as having the highest average budget overrun and Patel Construction as having the highest average delay within this dataset.

8. Project Status Analysis

Project Status	Projects	Share
Completed	504	50.4%
Ongoing	319	31.9%
Delayed	177	17.7%

Project status was analyzed using value counts and percentage distribution for Completed, Ongoing, and Delayed categories.

9. Relationship & Correlation Analysis

The analysis includes correlation checks between numeric project measures. One highlighted relationship is the approximately 0.89 positive correlation between project area and cost per square foot in the analysis. Correlation describes association in this dataset and does not by itself establish causation.

10. Visualizations Created

- Location-wise Actual Construction Cost — compares total actual cost across project locations.
- Average Budget Overrun by Project Type — compares average overrun percentages across project types.
- Delayed vs On-Time Projects — compares project counts by delay status.
- Estimated vs Actual Cost by Project Type — compares budget and actual spending by project type.
- Construction Cost Components — visualizes the share of Material, Labour, Equipment, and Other costs.
- Average Budget Overrun by Contractor — compares contractor-level average budget overrun.

11. Power BI Dashboard

The analysis is extended into an interactive Power BI dashboard with KPI cards, charts, slicers, and key-insight text. Dashboard KPIs include Total Projects, Total Estimated Budget, Total Actual Cost, Budget Overrun %, and Average Delay Days.

- Interactive filter: Location
- Interactive filter: Project Type
- Interactive filter: Contractor
- Interactive filter: Project Status

12. Key Analytical Tasks Covered

#	Task
1	Load CSV data into Pandas
2	Inspect dataset shape and structure
3	Check data types and non-null values
4	Generate descriptive statistics
5	Calculate total and average estimated budgets
6	Calculate total actual construction cost
7	Calculate budget variance and overrun percentage
8	Classify projects as under-budget or over-budget
9	Analyze overrun by project type
10	Analyze actual cost by location
11	Break total cost into major components
12	Analyze delayed vs on-time projects
13	Calculate average and maximum delay
14	Identify highly delayed projects
15	Analyze contractor performance metrics
16	Analyze project status distribution
17	Calculate status percentages
18	Analyze numeric relationships/correlation
19	Create analytical charts
20	Build KPI measures in Power BI
21	Build interactive Power BI dashboard
22	Summarize business insights

13. Portfolio Summary

This project demonstrates an end-to-end Data Analytics workflow: structured data ingestion, exploratory analysis, financial and operational KPI calculation, segmentation, visualization, dashboard development, and business insight generation. The dataset is synthetic and is intended for educational and portfolio purposes.